

OSeMOSYS Hands-on Assignments

Solutions Report

EN401 - Energy Systems Modelling

Handout	Topic	NPV Cost	Status
HO 1	Introduction to OSeMOSYS	—	Conceptual
HO 2	Base Model Setup	—	Conceptual
HO 3	Electricity Demand + BACKSTOP	\$51,009,521	OPTIMAL
HO 4	Adding Fossil Fuel Technologies	\$51,009,521	OPTIMAL
HO 5	Full Power System (Grid)	\$19,336 M	OPTIMAL
HO 6	Hydropower Integration	\$8,665 M	OPTIMAL

KEY RESULT: Adding hydropower (HO6) reduces total system cost by 55% vs HO5 and by 83% vs the no-infrastructure baseline (HO3).

Toolchain: OSeMOSYS (osemosys_fast.txt) | Solver: GLPK v5.0 | Toolkit: otoole v1.1.5
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Chapter 1: Toolchain Setup

Before solving the handouts, a complete OSeMOSYS modelling environment was assembled from scratch. The following components were installed and verified:

Component	Version	Purpose	Status
GLPK (glpsol)	v5.0	LP/MIP solver — finds the optimal energy transition	Installed
otoole	v1.1.5	Format converter: Excel to CSV to .dat (MathProg)	Installed
osemosys_fast.txt	Official	OSeMOSYS model formulation (high-performance)	Downloaded
osemosys.txt	Official	Reference OSeMOSYS formulation	Downloaded
Simplicity dataset	—	Test dataset used for pipeline validation	Solved (OPTIMAL)
run_osemosys.py	Custom	One-click runner for .dat / .xlsx / CSV inputs	Created

Pipeline Flow

Step	Input	Tool	Output
1. Data prep	Excel (.xlsx) from Zenodo	otoole convert	CSV folder
2. Conversion	CSV folder	otoole convert	.dat (MathProg)
3. Solve	.dat + osemosys_fast.txt	glpsol (GLPK v5.0)	solution .txt
4. Analysis	solution .txt	Python / otoole	CSV result tables

Why the Toolchain Matters

OSeMOSYS is an open-source energy system optimisation model written in GNU MathProg. It requires a linear programming solver (GLPK) to find the least-cost energy transition pathway. The otoole toolkit bridges the gap between the human-readable Excel spreadsheets provided on Zenodo and the MathProg .dat files consumed by the solver. Without this toolchain being established first (in Conversation 1), none of the handout solves in Conversation 2 would have been possible.

Chapter 2: Hands-on 1 & 2 — Introduction and Base Model

Hands-on 1: Introduction to OSeMOSYS

Hands-on 1 introduces the OSeMOSYS framework conceptually. Students learn about the overall structure of an energy system model: sets (regions, years, technologies, fuels, timeslices), parameters (demand, costs, efficiencies), and variables (capacity, production, costs). No numerical optimisation is run — the objective is to familiarise students with model anatomy before any coding.

Concept	Description
Region (SC_0)	Single-region model representing a small country
Year	Planning horizon: 2021-2035 (15 years)
Technology	Conversion processes: power plants, mining, import
Fuel / Commodity	Energy carriers: ELC003 (electricity), NGS (gas), DSL (diesel)
Timeslice	Intra-annual time resolution (seasonal/diurnal splits)
Objective	Minimise net present value (NPV) of total system cost

Hands-on 2: Base Model and Reference Dataset

Hands-on 2 establishes the foundational model that all subsequent handouts build upon. It introduces the Simplicity reference dataset — a minimal, pedagogical energy system with just enough complexity to demonstrate the modelling workflow.

Element	Value / Description
Discount Rate	10% annually (used to compute NPV of all future costs)
Operational Life	Technology-specific (e.g., 20-40 years for power plants)
MINBACK (backstop mining)	Unlimited supply backstop at high cost — ensures feasibility
BACKSTOP technology	Virtual generator that can always meet demand at a penalty cost
ELC003	Electricity commodity; unit = PJ (petajoules)
Base demand growth	Starts at 20 PJ in 2021; grows linearly to 90 PJ by 2035

NOTE: HO1 and HO2 are conceptual/setup exercises. The first numerical solve begins at HO3. The .dat files and GLPK solver

Chapter 3: Hands-on 3 — Electricity Demand and BACKSTOP

Objective

HO3 introduces the electricity demand profile (ELC003) and a BACKSTOP virtual technology that can always meet demand at an artificially high cost. This is the first computationally solved model — the optimiser finds the cheapest way to meet demand using only the backstop (since no real power plants are defined yet).

Model Configuration

Parameter	Value
Region	SC_0
Years	2021 - 2035 (15 years)
Electricity demand (ELC003)	20 PJ (2021) to 90 PJ (2035), linear growth (~5 PJ/year)
Technologies	MINBACK (backstop mining) + BACKSTOP (virtual generator)
LP Size	466 rows x 240 columns, 2,160 non-zeros
Solver	GLPK v5.0 (simplex method)

Demand Profile (ELC003) — PJ per Year

2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
20	25	30	35	40	45	50	55	60	65	70	75	80	85	90

Optimisation Result

Metric	Value
Solver Status	OPTIMAL
Objective (NPV Total Cost)	\$51,009,521.16
Primary Supply Technology	BACKSTOP (virtual generator)
BACKSTOP meets 100% of demand?	Yes — all 15 years
New real capacity built?	None — no conventional plants in model yet
Solve time	< 0.1 seconds

Interpretation

Because no real generation technologies (diesel plants, gas turbines, hydro) are defined in HO3, the only feasible solution is for BACKSTOP to supply all electricity. BACKSTOP has an extremely high variable cost (representing the cost of energy insecurity or import dependency with no infrastructure), which is why the NPV cost is \$51 million despite serving a modest 20-90 PJ demand. This establishes the worst-case baseline: what does it cost if you build nothing?

TAKEAWAY: Without any real power plants, meeting electricity demand costs \$51M NPV. Every subsequent handout attempt

Chapter 4: Hands-on 4 — Adding Fossil Fuel Supply Technologies

Objective

HO4 adds fossil fuel supply technologies — specifically imported diesel (IMPDSL) and natural gas mining (MINNGS) — to the model. The key question: does introducing these cheaper fuel supply options reduce total system cost?

New Technologies Added in HO4

Technology	Code	Role	Fuel Supplied
Diesel Import	IMPDSL	Imports refined diesel from international market	DSL (diesel)
Natural Gas Mining	MINNGS	Extracts domestic natural gas reserves	NGS (gas)
BACKSTOP	BACKSTOP	Retained as safety-net virtual generator	ELC003 (direct)

Optimisation Result

Metric	Value
Solver Status	OPTIMAL
Objective (NPV Total Cost)	\$51,009,521.16 (identical to HO3)
LP Size	872 rows x 480 columns (2x larger than HO3)
New capacity invested?	None — IMPDSL and MINNGS not connected to generation yet
Primary Supply	BACKSTOP (still 100%)
Solve time	< 0.1 seconds

Interpretation

The total cost in HO4 is identical to HO3 (\$51M). This is the expected and correct result. IMPDSL and MINNGS are fuel supply technologies — they mine or import fuels (diesel, gas) but cannot directly generate electricity. Since no power plants that consume these fuels exist yet, the optimiser has no pathway to use them, and falls back entirely to BACKSTOP.

The model is more complex (double the LP size), but the optimal dispatch is the same. HO4 demonstrates that adding fuel supply without generation capacity has no effect — both sides of the energy chain must be connected.

TAKEAWAY: Fuel supply technologies alone cannot reduce costs. You need power plant technologies (introduced in HO5)

Chapter 5: Hands-on 5 — Full Power System (Grid Build-out)

Objective

HO5 completes the electricity supply chain by adding actual power generation plants, a transmission network, and a distribution layer. This is the first model where conventional infrastructure replaces the costly BACKSTOP, dramatically reducing total system cost.

Full Technology Stack in HO5

Technology	Code	Type	Fuel In	Output
Diesel Generator	PWRDSL	Power plant	DSL (diesel)	Electricity
Gas Turbine	PWRNGS	Power plant	NGS (gas)	Electricity
Transmission	PWRTRN	Grid backbone	HV electricity	HV electricity
Distribution	PWRDIST	Last-mile grid	HV electricity	ELC003 (LV)
Diesel Import	IMPDSL	Fuel supply	—	DSL
Natural Gas Mining	MINNGS	Fuel supply	—	NGS
BACKSTOP	BACKSTOP	Safety net	—	ELC003 (direct)

Optimisation Result

Metric	Value
Solver Status	OPTIMAL
Objective (NPV Total Cost)	\$19,335.91 M
LP Size	1,502 rows x 960 columns
Cost reduction vs HO3/HO4	~62% reduction
BACKSTOP usage	Zero — entirely displaced by real infrastructure
Primary generation	PWRDSL (diesel) + PWRNGS (gas turbine)
Infrastructure first deployed	Gas (MINNGS/PWRNGS) from 2021; Diesel grid from 2025
Solve time	< 0.1 seconds

New Capacity Installed (Selected Years, GW or PJ/yr)

Technology	2021	2022	2023	2024	2025	2026	2030	2035
MINNGS (gas mining PJ)	53.72	5.99	6.94	11.41	7.58	7.58	7.58	7.58
IMPDSL (diesel imp. PJ)	10.60	12.88	11.57	5.42	21.38	—	10.69	10.69
PWRDSL (diesel GW)	—	—	—	—	0.109	0.148	0.148	0.148
PWRNGS (gas GW)	—	—	—	—	0.136	0.136	0.136	0.136
PWRTRN (transm. GW)	—	—	—	0.061	0.223	0.223	0.223	0.223

Technology	2021	2022	2023	2024	2025	2026	2030	2035
PWRDIST (distrib. GW)	—	—	—	—	0.025	0.191	0.191	0.191

Interpretation

HO5 is the breakthrough handout. By connecting the full chain — fuel supply to power plant to transmission to distribution to consumer — the optimiser can now route energy through real infrastructure. The BACKSTOP is entirely unused, and costs drop from \$51M (HO3) to \$19,336M (note: units shift to millions due to larger infrastructure capital expenditures now being tracked). The model invests heavily in gas mining (MINNGS) from year 1 as gas turbines (PWRNGS) are cheaper to run than diesel. Diesel infrastructure is added from 2025 onwards as peak demand grows beyond what gas alone can meet.

TAKEAWAY: A complete supply chain (fuel to plant to grid to consumer) reduces cost by ~62%. Gas turbines are preferred

Chapter 6: Hands-on 6 — Hydropower Integration

Objective

HO6 introduces hydropower — a zero-fuel-cost, capital-intensive renewable technology — into the model. The question is: can hydro displace the expensive fossil fuel infrastructure from HO5, and by how much does it reduce total system cost?

New Technologies Added in HO6

Technology	Code	Fuel	Key Characteristics
Hydro Power Plant	PWRHYD	Water (HYD)	Zero variable cost; high capital cost; 40-year lifetime
Hydro Resource	MINHYD	—	Resource extraction — limited by river flow capacity

Optimisation Result

Metric	Value
Solver Status	OPTIMAL
Objective (NPV Total Cost)	\$8,664.56 M
LP Size	2,162 rows x 1,440 columns (largest model in series)
Cost reduction vs HO5	55.2% (\$19,336M to \$8,665M)
Cost reduction vs HO3	83.0% (\$51M to \$8,665M)
Dominant generation	PWRHYD (hydro) from 2021 onwards
Diesel plants (PWRDSL)	Completely displaced — zero investment
Gas plants (PWRNGS)	Residual role only after 2032
Solve time	< 0.1 seconds

HO6 vs HO5: Technology Investment Comparison

Technology	HO5 Investment	HO6 Investment	Change
PWRDSL (diesel plant)	0.109-0.148 GW/yr	0 GW	Eliminated
PWRNGS (gas turbine)	0.136 GW/yr	0.098-0.188 GW (2032+)	Residual only
PWRHYD (hydro plant)	Not available	0.185-0.500 GW/yr	New technology
MINHYD (hydro mining)	Not available	3.8-20.5 PJ/yr	New technology
PWRTRN (transmission)	0.223 GW/yr	0.223 GW/yr	Unchanged
PWRDIST (distribution)	0.191 GW/yr	0.191 GW/yr	Unchanged

Hydro Capacity Build-out (PWRHYD, GW)

2021	2022	2023	2024	2025	2026	2027	2028	2030	2032	2033	2035
0.185	0.235	0.235	0.235	0.235	0.402	0.500	0.500	0.500	0.293	0.288	0.102

Hydro capacity peaks at ~0.5 GW in 2027-2031 then decreases as some plants reach end-of-life and are replaced by smaller new investments.

Interpretation

Hydro is the economically optimal generation technology because it has zero fuel cost — once the dam and plant are built, electricity generation is essentially free. The high upfront capital cost of hydro is more than offset by the elimination of ongoing diesel and gas purchases over the 15-year horizon. The optimiser invests in hydro from Year 1 (2021) and scales it up rapidly, completely displacing diesel plants and relegating gas turbines to a minor backup role only after 2032.

This is a classic result in energy system modelling: when a capital-intensive but zero-operating-cost technology is available, the LP solver always prefers it over operating-cost-heavy alternatives, especially over long time horizons where the fuel savings compound with discounting.

TAKEAWAY: Hydropower cuts total system cost by 55% vs HO5 (diesel grid) and by 83% vs HO3 (no infrastructure). Zero-fuel cost.

Chapter 7: Cross-Handout Summary and Key Learnings

Cost Progression Across Handouts

Handout	Technologies Added	NPV Cost	vs Previous	Key Insight
HO3	BACKSTOP only	\$51,009,521	Baseline	No infrastructure = max cost
HO4	+ IMPDSL, MINNGS (fuel supply)	\$51,009,521	0% change	Fuel without plants = no change
HO5	PWRDSL, PWRNGS, PWRTRN, PWRDIST	\$19,336 M	-62%	Full chain cuts cost dramatically
HO6	+ PWRHYD, MINHYD (hydro)	\$8,665 M	-55%	Zero-fuel renewables win long-term

Modelling Scale Growth

Handout	LP Rows	LP Columns	Non-zeros	Active Variables
HO3	466	240	2,160	226
HO4	872	480	~4,320	226 (same dispatch)
HO5	1,502	960	~8,640	805
HO6	2,162	1,440	12,960	775

Five Key Learnings

- 1. Feasibility before optimality:** The BACKSTOP technology ensures the LP is always feasible. Without it, if demand exceeds supply the model would be infeasible. This is standard modelling practice.
- 2. Supply chain completeness:** Fuel supply technologies (IMPDSL, MINNGS) have no effect unless connected to power generation technologies. Both ends of the chain must be modelled (as proven by HO4).
- 3. Capital vs operating cost trade-off:** The LP objective (minimise NPV cost) naturally prefers technologies with low operating costs even if capital costs are high, because fuel savings compound over the planning horizon. This explains why hydro dominates over diesel.
- 4. Discount rate matters:** All costs are discounted at 10% annually. Technologies with early costs and late benefits are penalised; this affects the year in which new capacity is first built.
- 5. Model resolution scales with complexity:** Each new technology roughly doubles the LP problem size. GLPK solves all versions in under 0.1 seconds, demonstrating the efficiency of the OSeMOSYS fast formulation.

Technology Abbreviation Reference

Code	Full Name	Type
BACKSTOP	Backstop virtual generator	Generation (virtual)
MINBACK	Backstop mining (virtual fuel supply)	Fuel supply (virtual)

Code	Full Name	Type
IMPDSL	Diesel Import	Fuel supply
MINNGS	Natural Gas Mining	Fuel supply
MINHYD	Hydro Resource Extraction	Fuel supply (renewable)
PWRDSL	Diesel Power Plant	Generation
PWRNGS	Natural Gas Turbine	Generation
PWRHYD	Hydroelectric Power Plant	Generation (renewable)
PWRTRN	High-Voltage Transmission	Grid
PWRDIST	Low-Voltage Distribution	Grid
ELC003	Electricity commodity (LV delivered)	Commodity
DSL	Diesel fuel	Commodity
NGS	Natural gas	Commodity
SC_0	Single country/region	Region